Quantum-Fuzzy Logical Petri Nets: A Formal Operator Framework for Hybrid Discrete, Fuzzy, and Quantum Dynamics

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Abstract

This paper develops a formal operator framework for Quantum-Fuzzy Logical Petri Nets (QFLPN). The central problem is the integration of discrete Petri-net structure, fuzzy activation, and quantum-state evolution without identifying fuzzy membership with quantum amplitude or measurement probability. The framework defines the QFLPN tuple, local and global quantum markings, bounded fuzzy activation, transition admissibility, firing semantics, operator construction, and quantum-state transformations. The global state space is constructed through tensor products of local Hilbert spaces, while closed-system evolution is represented by unitary operators and open-system evolution by completely positive trace-preserving maps. A reference mapping from a fuzzy activation degree to a rotation parameter is stated explicitly as a modelling convention. Analytical properties of density operators and unitary evolution are derived, and a deterministic four-qubit reference instance fixes the dimension, basis ordering, controlled-operation convention, and numerical verification conditions. The paper separates analytical consequences of the formalism from empirical claims that require dedicated experiments. The resulting framework is intended as a formal foundation for subsequent studies of sparse state evolution, matrix-function algorithms, degradation, decoherence, spectral stability, and comparative representations.

Keywords: quantum-fuzzy Petri nets; QFLPN; operator formalism; fuzzy activation; quantum state evolution; density operators; hybrid discrete-event systems

1 Introduction

Petri Nets provide a compact formal language for representing discrete-event structure, concurrency, precedence, and state change. Quantum Petri Net models extend this perspective toward quantum programs and quantum-state evolution, while evolutionary and hybrid quantum-centric models demonstrate that Petri-net abstractions can be used to connect computational structure with quantum operations [3, 4, 5]. The combination of discrete structure with graded activation introduces an additional semantic layer: a transition may be activated to a degree rather than through a purely Boolean enabling condition. A further quantum layer then requires a mathematically valid representation of the state and its transformation.

The resulting modelling problem is not solved by merely placing three labels on the same diagram. A Petri transition has a structural meaning, a fuzzy activation has a numerical meaning in an application-specific inference mechanism, and a quantum operator has an algebraic meaning on a Hilbert space. These meanings must remain distinct while a formally specified interface connects them. This requirement becomes particularly important when a scalar in $[0, 1]$ is mapped to a quantum rotation parameter. Such a scalar can be used to parameterize an operator, but it is not thereby redefined as a quantum amplitude or as a measurement probability.

The objective of this paper is to establish that interface as a formal operator framework. The framework is designed around five requirements. First, the Petri-net structure must be explicitly defined. Second, fuzzy activation must have a declared domain and range. Third, local quantum states must be associated with well-defined Hilbert spaces. Fourth, transition firing must specify the operator or quantum channel applied to the state. Fifth, the formal definitions must admit a finite reference instance that can be reproduced independently.

The resulting conceptual flow is

$$\begin{aligned} &\text{discrete marking} \longrightarrow \text{fuzzy activation} \longrightarrow \text{operator parameter} \\ &\text{operator parameter} \longrightarrow \text{quantum transformation} \longrightarrow \text{new marking.} \end{aligned} \tag{1}$$

The scientific contribution is the explicit semantic interface rather than a performance claim. The paper therefore does not infer large-scale speedup, numerical superiority, physical robustness, or experimental hardware performance from the formalism. Those questions require independent protocols and measured evidence. This separation follows standard scientific manuscript practice in which methodology, obtained results, interpretation, and limitations are kept distinguishable [6, 7].

2 Research Context and Problem Definition

2.1 Discrete-event and quantum modelling context

Quantum computation is commonly expressed through state vectors, density operators, unitary operators, and quantum channels. Petri-net formalisms provide a complementary structural representation in which places and transitions express state storage and events. Previous QPN research has used this representation for quantum programs, quantum evolutionary systems, and quantum-centric dynamic applications [3, 5, 4]. These developments motivate a framework in which a transition can carry both logical structure and a parameterized quantum transformation.

The QFLPN construction differs from a purely quantum Petri Net because the transition enabling mechanism contains an explicit fuzzy layer. Conversely, it differs from a fuzzy Petri Net because the state associated with a quantum-enabled place is not merely a scalar membership value. The formalism must therefore describe two coupled but non-identical state spaces: the discrete/fuzzy control space and the quantum state space.

2.2 Semantic gap

The central semantic gap can be stated precisely. Let $\mu \in [0, 1]$ be a fuzzy membership value and let $|\psi\rangle \in \mathcal{H}$ be a quantum state. There is no mathematical identity

$$\mu \equiv |\langle 1|\psi\rangle|^2 \tag{2}$$

without an explicitly defined encoding and state preparation. A QFLPN model must instead define a mapping

$$\Gamma : [0, 1] \to \mathcal{P}, \tag{3}$$

where $\mathcal{P}$ is a parameter space for admissible quantum operators. The particular choice of Γ is part of the model specification.

This distinction prevents a common category error. Fuzzy activation expresses graded logical or inferential relevance; quantum amplitudes are complex coefficients of a state vector; probabilities are obtained from a state and a measurement prescription. The three quantities can be related, but they are not interchangeable by notation alone.

2.3 Research question and scope

The research question is: how can a formalism combine Petri-net structure, fuzzy activation, and quantum-state evolution while preserving the semantic distinction between these layers and providing directly testable mathematical invariants?

The scope of P1 is the formal answer. The paper defines the model, transition semantics, operator interface, analytical properties, a reference four-qubit construction, and a reproducibility protocol. Large-scale sparse benchmarks, Taylor and Krylov comparisons, integral operators, degradation, decoherence, spectral convergence, tensor-network comparisons, and application-level validation are deliberately outside the evidentiary scope of this paper.

3 Formal QFLPN Architecture

3.1 QFLPN tuple

Let the Quantum-Fuzzy Logical Petri Net be represented by

$$\mathcal{N}_{\text{QFLPN}} = (P, T, E, \mathcal{H}, \mathcal{F}), \tag{4}$$

where P is the set of places, T is the set of transitions, E is the structural relation connecting places and transitions, $\mathcal{H}$ is the collection of Hilbert spaces associated with quantum-enabled components, and $\mathcal{F}$ is the fuzzy activation mechanism.

For a concrete instance, the structural relation can be represented by directed arcs with application-specific weights or labels. The abstract tuple intentionally does not force one arc semantics, because the same operator interface can be implemented with ordinary, weighted, inhibitor, timed, or application-specific relations provided that transition admissibility is defined explicitly.

3.2 Semantic layers

The architecture is separated into four operational layers. The structural layer contains the Petri-net topology and determines which places are connected to each transition. The fuzzy layer evaluates activation from the relevant memberships. The operator layer maps the activation and other parameters to an admissible quantum transformation. The state layer applies the transformation to the local or global quantum state.

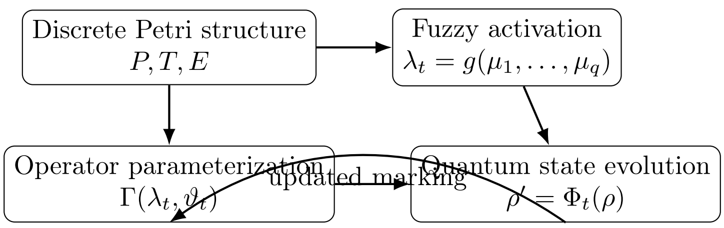

Figure 1: Semantic architecture of the QFLPN model. The four layers are coupled through explicit interfaces rather than by identifying their state variables.

The architecture is cyclic because the resulting quantum state may influence subsequent enabling or measurement conditions. The cycle does not imply that every implementation must measure the quantum state after every transition. It states only that the model admits feedback from state evolution to subsequent transition conditions when the application requires it.

3.3 Local and global quantum markings

For every quantum-enabled place $p \in P$, let $\mathcal{H}_p$ denote its local Hilbert space. The local quantum marking is represented by a density operator in

$$\mathcal{M}_Q(p) \in \mathcal{D}(\mathcal{H}_p). \tag{5}$$

For a composite system, the global Hilbert space is

$$\mathcal{H}_{\text{glob}} = \bigotimes_{p \in P} \mathcal{H}_p. \tag{6}$$

If the local dimensions are $d_p = \dim(\mathcal{H}_p)$, the global dimension is

$$D = \prod_{p \in P} d_p. \tag{7}$$

For q qubits, $d_p = 2$ for each qubit and therefore

$$D = 2^q. \tag{8}$$

This distinction is retained throughout the paper: q denotes the number of qubits, while D denotes the dimension of the associated Hilbert space.

3.4 Density-operator admissibility

A density operator ρ must satisfy

$$\rho \succeq 0, \qquad \rho = \rho^\dagger, \qquad \operatorname{Tr}(\rho) = 1. \tag{9}$$

The three constraints have different verification roles. Positive semidefiniteness guarantees non-negative probabilities for projective measurements. Hermiticity guarantees that expectation values of Hermitian observables are real. Unit trace normalizes the total probability.

These are formal admissibility conditions, not experimental observations. A numerical implementation can test their residuals, but a numerical test does not redefine the conditions themselves.

4 Fuzzy Activation and Operator Encoding

4.1 Activation function

For transition $t \in T$, let the relevant fuzzy memberships be $\mu_1(t), \ldots, \mu_q(t)$. The transition activation is

$$\lambda_t = g\big(\mu_1(t), \ldots, \mu_q(t)\big), \tag{10}$$

with the explicit requirement

$$0 \le \lambda_t \le 1. \tag{11}$$

The function g belongs to the fuzzy layer. It may represent an application-specific aggregation rule, but its definition must be supplied for each concrete implementation.

The bound on λ_t is important because the reference quantum encoding uses the square root of the activation. If a computational implementation produces a value outside the interval, the model must either reject the transition or explicitly define a normalization or saturation mechanism. Silent clipping would change the semantics and should not be introduced without being part of the model.

4.2 Reference activation-to-rotation map

For the reference QFLPN encoding, the activation is mapped to a rotation angle according to

$$\theta_t = 2 \arcsin \sqrt{\lambda_t}. \tag{12}$$

The corresponding single-qubit rotation is

$$R_Y(\theta_t) = \begin{bmatrix} \cos(\theta_t/2) & -\sin(\theta_t/2) \\ \sin(\theta_t/2) & \cos(\theta_t/2) \end{bmatrix}. \tag{13}$$

For the initial state $|0\rangle$,

$$R_Y(\theta_t)|0\rangle = \cos(\theta_t/2)|0\rangle + \sin(\theta_t/2)|1\rangle. \tag{14}$$

Consequently, under this specific preparation and measurement convention,

$$\Pr(1) = \sin^2(\theta_t/2) = \lambda_t. \tag{15}$$

This equality is a property of the declared encoding and initial state; it is not a definition of fuzzy membership in general.

4.3 Alternative encodings

The formalism permits an alternative mapping

$$\Gamma : [0,1] \times \Theta \to \mathcal{U}, \tag{16}$$

where Θ denotes additional classical parameters and $\mathcal{U}$ is the set of admissible unitary operators. An alternative encoding is acceptable only if its domain, range, state preparation, operator action, and verification conditions are declared. The existence of alternatives is useful for future comparative studies but does not weaken the status of the reference mapping used in P1.

5 Transition Admissibility and Firing Semantics

5.1 Structural enabling

A transition $t \in T$ is structurally admissible when the marking satisfies the input conditions associated with its incoming arcs. Denote the structural predicate by

$$A_{\mathrm{str}}(t, M) = 1. \tag{17}$$

The exact predicate depends on the Petri-net interpretation of the arcs and tokens. The essential requirement is that the structural condition be evaluated before the quantum operator is applied.

5.2 Fuzzy enabling

The fuzzy condition is represented by

$$A_{\mathrm{fz}}(t, M) = \lambda_t, \qquad \lambda_t \in [0,1]. \tag{18}$$

A thresholded implementation may introduce a declared parameter $\tau \in [0,1]$ and define an admissibility predicate

$$A_\tau(t, M) = \mathbf{1}_{\{\lambda_t \geq \tau\}}. \tag{19}$$

The threshold is optional. If it is not part of the application, the activation remains a graded parameter rather than a Boolean firing condition.

5.3 Quantum transformation

For a closed quantum subsystem, transition firing applies a unitary operator U_t:

$$\rho' = U_t \rho U_t^\dagger, \qquad U_t^\dagger U_t = I. \tag{20}$$

For an open subsystem, the more general representation is a completely positive trace-preserving map

$$\rho' = \Phi_t(\rho). \tag{21}$$

A Kraus representation can be written as

$$\Phi_t(\rho) = \sum_a K_{t,a} \rho K_{t,a}^\dagger, \tag{22}$$

with

$$\sum_a K_{t,a}^\dagger K_{t,a} = I. \tag{23}$$

This extension is essential when later studies include noise, decoherence, or degradation. P1 defines the interface but does not claim measured decoherence performance.

5.4 Partial observation and subsystem evolution

When only a subsystem is observed, the reduced state is obtained by the partial trace. For a bipartite state ρ_{AB},

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}). \tag{24}$$

This operation permits a QFLPN model to maintain a global state while exposing only the information required by a classical controller or another transition. The distinction between the global state and a reduced state must remain explicit because entanglement can make a reduced state mixed even when the global state is pure.

6 Operator Semantics and Analytical Properties

6.1 Unitary preservation of trace

For $\rho' = U\rho U^\dagger$ and unitary U,

$$\mathrm{Tr}(\rho') = \mathrm{Tr}(U\rho U^\dagger) = \mathrm{Tr}(U^\dagger U \rho) = \mathrm{Tr}(\rho) = 1. \tag{25}$$

Thus trace normalization is analytically preserved.

6.2 Hermiticity preservation

If $\rho = \rho^\dagger$, then

$$(U\rho U^\dagger)^\dagger = U\rho^\dagger U^\dagger = U\rho U^\dagger. \tag{26}$$

Hermiticity is therefore preserved exactly by unitary evolution.

6.3 Positive semidefiniteness preservation

For any vector $|v\rangle$, define $|w\rangle = U^\dagger |v\rangle$. Then

$$\langle v|\rho'|v\rangle = \langle v|U\rho U^\dagger|v\rangle = \langle w|\rho|w\rangle \geq 0, \tag{27}$$

provided $\rho \succeq 0$. Hence positive semidefiniteness is preserved.

6.4 CPTP preservation

For the open-system map Φ_t, complete positivity and trace preservation imply that valid density operators are mapped to valid density operators. The trace-preserving property follows directly from the Kraus completeness relation:

$$\mathrm{Tr}(\Phi_t(\rho)) = \sum_a \mathrm{Tr}(K_{t,a}\rho K_{t,a}^\dagger) = \mathrm{Tr}\left(\rho \sum_a K_{t,a}^\dagger K_{t,a}\right) = \mathrm{Tr}(\rho). \tag{28}$$

The complete-positivity requirement concerns positivity under extension with arbitrary ancillary systems and is the appropriate mathematical condition for a physical quantum channel.

6.5 Observable probabilities

For a projective measurement with projector Π, the probability is

$$p(\Pi) = \mathrm{Tr}(\Pi\rho). \tag{29}$$

The expression demonstrates why measurement probability belongs to the quantum layer. It depends on both the state and the measurement operator. A fuzzy membership value does not become a measurement probability unless the QFLPN instance explicitly constructs an encoding for which the equality holds.

7 Transition Semantics and Formal Execution

The firing semantics are defined as a composition of structural, fuzzy, and quantum maps. Let $M = (M_C, \rho)$ denote the complete QFLPN marking, where M_C is the discrete component and ρ is the quantum component. For a transition t, define the structural admissibility predicate $A_{\mathrm{str}}(t, M_C)$ and the fuzzy activation λ_t.

The transition is structurally admissible when

$$A_{\mathrm{str}}(t, M_C) = 1. \tag{30}$$

The fuzzy stage then evaluates

$$\lambda_t = g\big(\mu_1(t), \dots, \mu_q(t)\big), \qquad 0 \le \lambda_t \le 1. \tag{31}$$

The operator parameter is obtained through the declared encoding

$$\vartheta_t = \Gamma(\lambda_t, \xi_t), \tag{32}$$

where ξ_t collects additional classical parameters. The corresponding quantum transformation is

$$\rho' = \Phi_t(\rho; \vartheta_t). \tag{33}$$

For a closed subsystem, $\Phi_t(\rho;\vartheta_t) = U_t(\vartheta_t)\rho U_t(\vartheta_t)^\dagger$. For an open subsystem, Φ_t is a CPTP map.

The complete marking update can therefore be written abstractly as

$$M' = F_t(M), \tag{34}$$

with

$$F_t(M_C, \rho) = \left(F_t^C(M_C), \Phi_t(\rho; \vartheta_t)\right), \tag{35}$$

subject to structural admissibility and the declared fuzzy domain constraints. This formulation makes the interface explicit: the Petri-net marking is updated by the structural semantics, while the quantum state is updated by the operator semantics.

7.1 Composed transitions

If a transition contains several local quantum operations acting on disjoint subsystems, the operator can be constructed as a tensor product,

$$U_t = U_{t,1} \otimes U_{t,2} \otimes \cdots \otimes U_{t,r}. \tag{36}$$

When the transition contains a controlled interaction, the operator is not factorized and must be specified directly or through a circuit decomposition. The formalism therefore accommodates both independent and coupled quantum actions without changing the Petri-net interface.

7.2 Sequential composition

For two admissible transitions t_a and t_b, sequential execution is represented by

$$\rho_2 = \Phi_{t_b}(\Phi_{t_a}(\rho_0)). \tag{37}$$

In the closed-system case this becomes

$$\rho_2 = U_{t_b} U_{t_a} \rho_0 U_{t_a}^{\dagger} U_{t_b}^{\dagger}. \tag{38}$$

The order is mathematically relevant whenever U_{t_a} and U_{t_b} do not commute. This provides a precise link between Petri-net precedence and non-commutative quantum operator composition.

7.3 Reachability representation

Let $\mathcal{R}(M_0)$ denote the set of markings reachable from an initial marking M_0 through admissible transition sequences. A reachable execution path has the form

$$M_0 \xrightarrow{t_1} M_1 \xrightarrow{t_2} M_2 \xrightarrow{t_3} \cdots \xrightarrow{t_k} M_k. \tag{39}$$

Unlike a finite Boolean Petri-net state graph, the QFLPN reachability space may contain continuous quantum states and continuously varying fuzzy parameters. Therefore, a numerical implementation should distinguish logical reachability from sampled numerical trajectories. The former is a semantic property of the model; the latter is an experimental or computational observation.

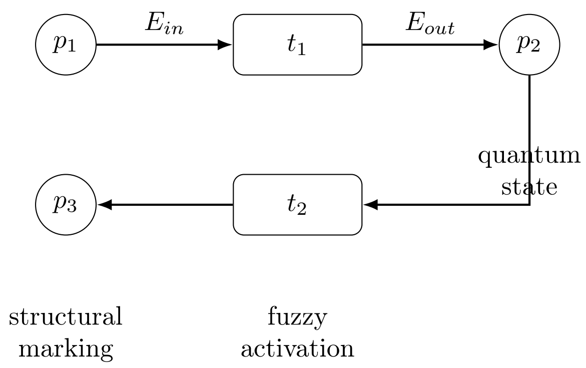

Figure 2: Minimal structural QFLPN motif. The Petri topology determines precedence, while the transition interface carries the fuzzy activation and quantum transformation associated with firing.

8 Structural Properties and Propositions

8.1 Proposition on closed-system state validity

Proposition 1. If ρ is a valid density operator and U is unitary, then $\rho' = U\rho U^\dagger$ is a valid density operator.

Proof. Hermiticity follows from

$$(U\rho U^\dagger)^\dagger = U\rho U^\dagger. \tag{40}$$

Trace preservation follows from cyclicity of the trace and $U^\dagger U = I$. Positive semidefiniteness follows because, for every vector $|v\rangle$, the quadratic form $\langle v|U\rho U^\dagger|v\rangle$ equals $\langle w|\rho|w\rangle$ for $|w\rangle = U^\dagger|v\rangle$. Therefore all three density-operator conditions are preserved. □

8.2 Proposition on sequential unitary composition

Proposition 2. If U_a and U_b are unitary, then the sequential transition operator U_bU_a is unitary.

Proof.

$$(U_bU_a)^\dagger(U_bU_a) = U_a^\dagger U_b^\dagger U_bU_a = U_a^\dagger U_a = I. \tag{41}$$

Thus the composition remains an admissible closed-system quantum transformation. □

8.3 Proposition on the reference encoding

Proposition 3. For $\lambda \in [0, 1]$, the reference parameter $\theta = 2 \arcsin \sqrt{\lambda}$ is real and satisfies $0 \leq \theta \leq \pi$.

Proof. Since $0 \leq \lambda \leq 1$, one has $0 \leq \sqrt{\lambda} \leq 1$. Therefore $\arcsin \sqrt{\lambda} \in [0, \pi/2]$, giving $\theta \in [0, \pi]$. □

These propositions establish formal properties that do not depend on a particular software implementation. They should be distinguished from numerical tests, which quantify floating-point residuals for a concrete implementation.

9 QFLPN Structural Specification

The formal definitions are summarized in Table 1. The purpose of this table is to make the model directly implementable: every object has a mathematical domain, a semantic role, and an evidence type. This is consistent with the modelling style of the supplied professor example, where Petri structure, place types, transition parameters, quantum operators, and reachability are specified before experimental execution.

10 Reference Four-Qubit Instance

10.1 Dimension and basis convention

The reference instance contains four qubits, hence

$$q = 4, \qquad D = 2^4 = 16. \tag{42}$$

The computational basis is ordered lexicographically from $|0000\rangle$ to $|1111\rangle$. The state representation therefore contains 16 complex amplitudes in the pure-state vector representation or a 16×16 density matrix in the mixed-state representation.

Table 1: Formal specification of the principal QFLPN objects.

Object	Mathematical form	Semantic role
Net	$\mathcal{N}_{QFLPN} = (P, T, E, \mathcal{H}, \mathcal{F})$	Defines the complete model and its domains
Structural relation	$E \subseteq (P \times T) \cup (T \times P)$	Encodes precedence and connectivity
Classical marking	M_C	Stores discrete or classical state information
Fuzzy activation	$\lambda_t = g(\mu_1, \ldots, \mu_q)$	Represents graded transition activation
Local quantum state	$\rho_p \in \mathcal{D}(\mathcal{H}_p)$	Represents a quantum subsystem
Global state	$\rho \in \mathcal{D}(\mathcal{H}_{glob})$	Represents the composite quantum system
Operator map	$\Gamma(\lambda_t, \xi_t)$	Converts declared classical parameters to quantum parameters
Transition map	F_t	Couples structural and quantum state updates
Reachability	$\mathcal{R}(M_0)$	Describes markings accessible through admissible firings

The basis convention is part of the reproducibility specification. A different tensor-product ordering may produce a different matrix layout even when the logical circuit is equivalent. The implementation must therefore document the qubit order before numerical outputs are interpreted.

10.2 Activation-controlled rotations

For each activation λ_k, the reference construction uses

$$\theta_k = 2 \arcsin \sqrt{\lambda_k} \tag{43}$$

and applies the corresponding $R_Y(\theta_k)$ operation to the declared qubit. The complete operator is constructed as a tensor product of local operators when the operations act independently, or as a controlled operator when the transition semantics require conditional coupling.

10.3 Controlled three-qubit condition

For a three-control operation, define

$$P_c = |111\rangle\langle 111|. \tag{44}$$

The controlled-X operator acting on the fourth qubit is

$$C^3X = (I_8 - P_c) \otimes I_2 + P_c \otimes X. \tag{45}$$

Because P_c and $I_8 - P_c$ are orthogonal projectors that sum to I_8, the operator applies X exactly on the $|111\rangle$ control subspace and the identity elsewhere.

10.4 Reference state-space table

The table is not an experimental result. It is a fixed specification that allows two implementations to be compared under the same mathematical conventions.

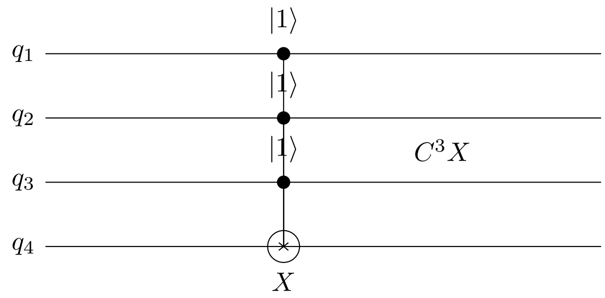

Figure 3: Logical representation of the three-control operation used by the four-qubit reference construction.

Table 2: Fixed conventions of the QFLPN four-qubit reference instance.

Property	Reference convention
Number of qubits	$q = 4$
Hilbert-space dimension	$D = 16$
Basis	Computational basis, lexicographic order
State-vector precision	Double precision
Single-qubit encoding	$\theta = 2 \arcsin \sqrt{\lambda}$
Reference rotation	$R_Y(\theta)$
Controlled operation	C^3X
Control projector	$P_c = \|111\rangle\langle 111\|$
State representation	Pure vector or density matrix, as declared
Verification	Trace, Hermiticity, positivity, dimension

11 Scalability and Instance Hierarchy

The four-qubit construction is a reference instance, not a scalability limit. The formalism is parameterized by the number of quantum subsystems and does not impose $q = 4$. The value $q = 4$ is selected because it is large enough to expose non-trivial controlled interaction while remaining small enough for exact, transparent verification of every matrix element. A scalable research program must distinguish this verification role from the computational regime used to study performance.

For an exact state-vector representation, the Hilbert-space dimension is

$$D = 2^q, \tag{46}$$

and a dense complex-double state vector requires approximately $16\,2^q$ bytes, before auxiliary storage and implementation overhead are included. Consequently, the computational difficulty is governed exponentially by q for generic exact state-vector simulation. For example, $q = 20$ corresponds to $D = 1,048,576$ amplitudes and approximately 16 MiB for the state vector alone, whereas $q = 30$ corresponds to approximately 16 GiB and $q = 40$ to approximately 16 TiB. These figures are representation-level estimates, not measured QFLPN results.

The hierarchy also clarifies why a statement such as "10,000 qubits" cannot be treated as a generic continuation of the four-qubit experiment. For a fully generic dense state-vector model, $q = 10{,}000$ would require $2^{10{,}000}$ amplitudes and is therefore not an exact classical state-vector target. Large qubit counts become meaningful only after the representation and the structure being exploited are specified. Tensor-network methods, sparse representations, stabilizer methods, symmetry reductions, or other compressed descriptions can access substantially larger structured systems, but their achievable size depends on entanglement, topology, circuit depth, observable, approximation

Table 3: Illustrative hierarchy of quantum-system sizes for exact state-vector representation.

q	2^q	Approx. state-vector storage	Scientific role
4	16	256 B	exact reference and semantic verification
8	256	4 KiB	transition-composition tests
12	4096	64 KiB	controlled-interaction tests
16	65536	1 MiB	small numerical benchmark
20	1,048,576	16 MiB	moderate exact benchmark
30	1,073,741,824	16 GiB	large exact state-vector regime
40	1,099,511,627,776	16 TiB	memory-dominated regime

tolerance, and contraction strategy. Recent benchmarking literature explicitly shows that simulator choice and circuit structure can change feasibility by more than an order of magnitude, while tensor-network approaches have their own representation-dependent limits.[1, 2]

This distinction is essential for the subsequent article series. Quantum-system size q, Petri-net size $(|P|, |T|)$, sparse matrix dimension N, and number of experimental repetitions are different quantities and must never be reported as interchangeable "system size". The P1 reference instance therefore remains four-qubit, while P2 and later studies may use larger quantum instances when the selected representation and available evidence support them. A larger instance is scientifically valuable only when it tests a new hypothesis rather than merely increasing a numerical parameter.

12 Formal Completeness and Execution Contract

The formal definitions above can be assembled into an execution contract that determines what must be specified before a QFLPN transition is considered mathematically well-defined. This contract is useful because the same Petri topology can otherwise admit different fuzzy aggregation rules, operator parameterizations, basis orderings, and measurement conventions. The model is therefore treated as a typed composition rather than as an informal correspondence between diagram elements.

For a transition t, define the semantic contract

$$\mathfrak{C}_t = (A_{\mathrm{str}}, g, \Gamma, \Phi, \mathsf{M}), \tag{47}$$

where A_{str} is the structural admissibility predicate, g is the fuzzy aggregation rule, Γ is the activation-to-operator parameter map, Φ is the quantum transformation, and M is the declared measurement or observation convention. A transition specification is complete only when the domains and codomains of these components are fixed.

For the closed-system reference case, the contract can be expressed as

$$M \xrightarrow{A_{\mathrm{str}}} \lambda_t \xrightarrow{\Gamma} \vartheta_t \xrightarrow{U_t(\vartheta_t)} \rho' \xrightarrow{\mathsf{M}} y, \tag{48}$$

where y denotes an optional classical observation. Each arrow has a distinct mathematical type. In particular, λ_t is a scalar activation, ϑ_t is an operator parameter, U_t is an operator, and y is an observation produced only after a measurement or declared readout operation.

The contract also fixes the direction of dependence. The fuzzy value may parameterize a quantum operator, but the quantum state is not altered merely by computing λ_t. Conversely, a quantum measurement may influence a subsequent classical marking only when the model explicitly includes a feedback rule. This distinction prevents an implicit information transfer between semantic layers.

12.1 Well-posed transition conditions

A transition is well-posed when the following conditions hold simultaneously: the structural predicate is defined on the current marking; the fuzzy aggregation produces a value in $[0, 1]$; the parameter map produces an element of its declared parameter space; and the quantum transformation maps admissible states to admissible states. For the unitary reference case this gives

$$A_{\text{str}}(t, M_C) = 1, \qquad 0 \le \lambda_t \le 1, \qquad U_t^\dagger U_t = I. \tag{49}$$

These conditions separate three different failure modes. A structural failure means that the transition is not enabled. A semantic failure means that the fuzzy or parameterization layer produced an invalid value. An operator failure means that the declared transformation does not satisfy the admissibility conditions required by the selected quantum semantics. Treating these cases separately makes implementation diagnostics scientifically interpretable.

For the reference encoding $\theta_t = 2 \arcsin \sqrt{\lambda_t}$, the activation constraint alone guarantees

$$0 \le \theta_t \le \pi. \tag{50}$$

The angle is therefore real and lies in the principal interval used by the stated single-qubit rotation convention. This is an analytical consequence of the encoding and does not require a numerical experiment.

12.2 Invariants under repeated firing

If a sequence of closed-system transitions is represented by unitary operators $U_1, \ldots, U_k$, the resulting state is

$$\rho_k = U_k \cdots U_2 U_1 \rho_0 U_1^\dagger U_2^\dagger \cdots U_k^\dagger. \tag{51}$$

Because the product $U_k \cdots U_1$ is unitary, density-operator validity is preserved for the entire sequence. Thus trace normalization, Hermiticity, and positive semidefiniteness are invariants of repeated closed-system firing.

This invariant statement is stronger than checking one transition in isolation. It establishes that the local correctness condition composes over an arbitrary finite sequence, provided every transition in that sequence satisfies the declared unitary contract. For open-system transitions, the analogous statement follows from closure of CPTP maps under composition.

12.3 Ordering, concurrency, and non-commutativity

Petri-net concurrency does not automatically imply quantum-operator commutativity. Consider two transitions t_a and t_b with operators U_a and U_b. If

$$[U_a, U_b] = U_a U_b - U_b U_a = 0, \tag{52}$$

then the corresponding closed-system state transformations are order-independent at the operator level. If instead

$$[U_a, U_b] \ne 0, \tag{53}$$

the execution order is semantically relevant and the two transition sequences generally produce different states.

For operations acting on disjoint tensor factors, commutativity follows from the tensor-product structure. If U_a acts only on subsystem A and U_b only on subsystem B, then

$$(U_a \otimes I_B)(I_A \otimes U_b) = (I_A \otimes U_b)(U_a \otimes I_B). \tag{54}$$

This provides a precise mathematical criterion for identifying a class of QFLPN transitions that can be treated as independent concurrent actions. Controlled interactions, shared subsystems, and overlapping supports require the commutator or an equivalent operator relation to be evaluated explicitly.

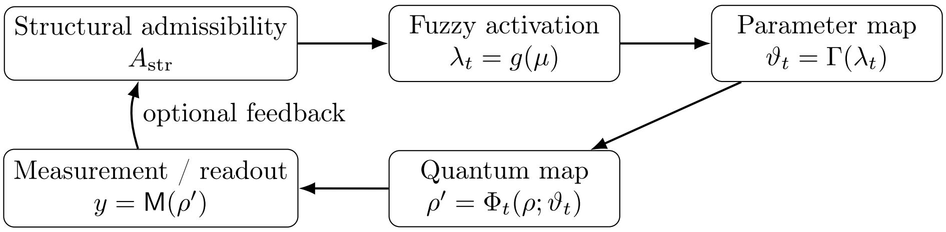

Figure 4: Execution contract for a QFLPN transition. Feedback from a measurement to structural enabling is optional and must be explicitly defined by the model.

12.4 Measurement and classical feedback

Let
Π_y be a projective measurement satisfying

$$\Pi_y = \Pi_y^\dagger, \qquad \Pi_y \Pi_z = 0 \quad (y \neq z), \qquad \sum_y \Pi_y = I. \tag{55}$$

The probability of outcome y is

$$p_y = \mathrm{Tr}(\Pi_y \rho). \tag{56}$$

If the QFLPN model uses this outcome to update a classical marking, a feedback function must be supplied explicitly, for example

$$M'_C = G(M_C, y). \tag{57}$$

The measurement therefore creates a well-defined interface from the quantum layer back to the discrete layer. Without a declared G, no classical state update should be inferred from the measurement merely because the measurement has been computed.

The same principle applies to partial observations. If $\rho_A = \mathrm{Tr}_B(\rho_{AB})$, then any controller using ρ_A must specify which functional of ρ_A is used for the next decision. This keeps the information flow auditable and prevents a reduced-state calculation from being mistaken for a full-state observation.

12.5 Formal verification obligations

The mathematical model yields a compact set of verification obligations that can be tested independently of performance experiments. The structural obligation checks the declared input and output relation of each transition. The fuzzy obligation checks the domain and range of the activation function. The operator obligation checks unitarity for closed transitions or the CPTP conditions for open transitions. The state obligation checks density-operator admissibility after each firing. The interface obligation checks that basis ordering, subsystem support, and measurement conventions are identical between the mathematical specification and the implementation.

These obligations define what can be established analytically before any large-scale numerical benchmark is attempted. They also explain why a small deterministic instance is scientifically useful: it can validate the complete semantic chain without conflating correctness with computational performance.

13 Computational Representation and Complexity

13.1 Dense representation

For a D-dimensional pure state, a state vector requires D complex amplitudes. A dense density matrix requires D^2 complex entries. For the four-qubit reference, $D = 16$, so the dense state-vector representation contains 16 amplitudes and the density-matrix representation contains 256 complex entries. These quantities are exact consequences of the representation dimension.

13.2 Sparse representation

When the operator or state exhibits exploitable sparsity, a sparse representation can reduce storage and arithmetic cost. For a matrix in compressed sparse row representation, the cost of one sparse matrix-vector multiplication is proportional to the number of stored nonzero elements, conventionally denoted by NNZ, together with lower-order indexing operations. This observation motivates the separate computational study of CSR and sparse evolution in P3.

13.3 Complexity boundary

The QFLPN tuple does not imply one universal computational complexity. Complexity depends on the state representation, operator structure, sparsity, algorithm, precision, hardware, and stopping criteria. A formal statement such as

$$\text{QFLPN is scalable} \tag{58}$$

would therefore be scientifically incomplete without specifying the computational regime and measured evidence. P1 instead establishes the representation-level quantities from which such analyses can later be derived.

14 Verification Framework

14.1 Analytical verification

Analytical verification establishes properties that follow from the definitions. Examples include the preservation of trace, Hermiticity and positive semidefiniteness under unitary evolution, and trace preservation under a CPTP map. These results are independent of a particular programming language or processor.

14.2 Numerical verification

Numerical verification evaluates a concrete implementation. The following residuals are directly relevant:

$$\varepsilon_{\mathrm{tr}} = |\operatorname{Tr}(\rho) - 1|, \tag{59}$$

$$\varepsilon_{\mathrm{H}} = \|\rho - \rho^\dagger\|, \tag{60}$$

and a positive-semidefiniteness diagnostic based on the minimum eigenvalue,

$$\varepsilon_{\mathrm{PSD}} = \max\{0, -\lambda_{\min}(\rho)\}. \tag{61}$$

The matrix norm used for ε_{H} and the numerical tolerance must be declared by the implementation. A result is reproducible only when these conventions are fixed.

14.3 Independent implementation check

A stronger verification protocol uses two independent implementations with the same input parameters and basis ordering. Agreement of the final state or declared observables within a pre-specified tolerance provides evidence of implementation consistency. It does not prove physical correctness in an experimental processor, but it substantially reduces the risk of a hidden implementation-specific error.

Table 4: Verification layers for the formal QFLPN framework.

Layer	Object tested	Evidence type
Formal	Tuple, domains, codomains	Definition and derivation
Operator	Unitarity or CPTP conditions	Algebraic verification
State	Trace, Hermiticity, positivity	Analytical and numerical residuals
Reference instance	Dimension and basis ordering	Deterministic specification
Implementation	Independent output agreement	Reproducible numerical test
Physical system	Hardware execution and noise	Experimental evidence

The last row is intentionally outside the present evidence boundary. A software-level formalism should not be presented as experimental hardware validation.

15 Reference Execution Protocol

The reproducibility protocol fixes the sequence of operations rather than prescribing an unsupported performance result. The implementation first declares the number of qubits and the tensor-product ordering. It then constructs the initial state and the fuzzy inputs. Each transition computes the activation, verifies its range, constructs the declared operator, applies the operator, and records the resulting state or measurement statistics. Verification residuals are recorded after each declared checkpoint.

For a finite reference experiment, the protocol should report the input activations, initial state, operator sequence, precision, basis ordering, measurement convention, software version, and numerical tolerances. If random sampling is introduced, the random seed and sampling count must also be recorded. If an external simulator or quantum processor is used, its version and execution configuration must be reported.

The protocol is intentionally more restrictive than a simple demonstration. A reproducibility claim concerns the ability of an independent researcher to reconstruct the same computational object, not merely to obtain a visually similar circuit.

16 Figures, Structural Objects, and Article Evidence

The formal paper benefits from structural objects that carry scientific information rather than decorative content. Figure 1 defines the semantic pipeline. Figure 3 fixes the logical structure of the controlled reference operation. Table 2 fixes the reference conventions, while Table 4 separates analytical, numerical, implementation, and physical evidence.

This organization follows the principle that figures and tables should expose exact values, relationships, or structures that are difficult to communicate as prose alone. It also keeps the main text free of editorial markers, development-stage labels, and left-margin numbering that do not belong to a scientific article. The document is therefore written as a normal article rather than as a project checklist.

17 Comparative Formal Positioning

The contribution of QFLPN is best assessed by comparing semantic responsibilities rather than by counting notational similarities. A conventional Petri-net model specifies discrete structure and firing relations. A fuzzy Petri-net extension introduces graded activation or membership. Quantum Petri-net formalisms introduce quantum states, amplitudes, operators, or measurement semantics. A QFLPN model is intended to preserve these roles while making the interfaces between them explicit.

Table 5 summarizes the distinction. The comparison is deliberately structural: it does not claim that one modelling family is universally superior to another, and it does not substitute for the dedicated reachability, performance, numerical, or representation comparisons that require their own experimental evidence.

Table 5: Formal positioning of QFLPN relative to adjacent modelling families.

Property	Classical Petri nets	Fuzzy Petri nets	Quantum Petri nets	QFLPN
Discrete places and transitions	explicit	explicit	explicit or model-dependent	explicit
Graded activation	usually absent	explicit	not intrinsic	explicit
Quantum state space	absent	absent	explicit	explicit
Activation semantics	Boolean enabling	membership rule	model-dependent	typed activation
Quantum transformation	absent	absent	operator dependent	declared operator or channel
Measurement semantics	absent	absent	quantum-layer dependent	explicit and separated from activation
Local-to-global state composition	structural only	structural only	tensor-product dependent	tensor-product construction
Open-system evolution	not intrinsic	not intrinsic	model-dependent	CPTP-compatible interface
Main scientific role in P1	baseline structure	activation semantics	quantum layer	integrated formal interface

The essential distinction is therefore not the presence of the three vocabularies but the existence of a typed semantic contract between them. In QFLPN, a fuzzy value is an input to a declared encoding function; the encoded parameter determines an operator; and the operator acts on a quantum state. This sequence prevents the common but invalid shortcut of treating a membership value as though it were automatically a complex amplitude or a measurement probability.

The comparison also clarifies what P1 does not establish. It does not establish a new reachability algorithm, a computational speedup, a superior matrix-function approximation, a decoherence advantage, or a tensor-network advantage. Those are independent claims requiring independent evidence. The value of P1 is the formal interface that makes such later claims precisely stateable and testable.

18 Compositionality and Semantic Consistency

A useful formalism should remain well-defined when several valid transitions are composed. This requirement is stronger than showing that an individual transition is mathematically admissible: the composition must preserve the declared state space and must respect the support of each operation.

18.1 Typed transition contract

For a transition t, let the semantic contract be written as

$$\mathfrak{C}_t = (A_t, g_t, \Gamma_t, \Phi_t, \mathfrak{M}_t), \tag{62}$$

where A_t denotes the structural admissibility relation, g_t the fuzzy activation rule, Γ_t the activation-to-parameter encoding, Φ_t the quantum transformation, and $\mathfrak{M}_t$ the declared observation convention. A transition is well-formed only when these components have compatible domains and codomains.

For an enabled transition, the semantic flow is therefore

$$M \longrightarrow \lambda_t \longrightarrow \vartheta_t \longrightarrow \Phi_t(\rho) \longrightarrow y_t, \tag{63}$$

where M denotes the relevant marking information, $\lambda_t \in [0, 1]$ is the fuzzy activation, ϑ_t is the encoded operator parameter, ρ is the quantum state, and y_t denotes the declared observable or measurement output. Each arrow represents a distinct semantic transformation rather than a change of notation for the same quantity.

18.2 Compositional proposition

Proposition. Let $t_1, \ldots, t_k$ be well-formed closed-system transitions whose quantum transformations are unitary on their declared supports. If each transition is structurally enabled and its activation-to-operator encoding produces a valid unitary transformation, then the sequential composition

$$U_{1:k} = U_k \cdots U_2 U_1 \tag{64}$$

is unitary on the corresponding global Hilbert space.

Proof. Since $U_j^\dagger U_j = I$ for every j, one obtains

$$U_{1:k}^\dagger U_{1:k} = U_1^\dagger \cdots U_k^\dagger U_k \cdots U_1 = U_1^\dagger \cdots I \cdots U_1 = I. \tag{65}$$

Thus the sequential composition is unitary. Consequently, if ρ is a valid density operator, then $U_{1:k}\rho U_{1:k}^\dagger$ remains positive semidefinite, Hermitian, and trace one. The proposition concerns mathematical admissibility only; it makes no claim about numerical conditioning or physical implementation error.

18.3 Support-aware concurrency

Suppose two transition operators act on disjoint tensor factors, with

$$U_a = U_a^{(S_a)} \otimes I_{S_b}, \qquad U_b = I_{S_a} \otimes U_b^{(S_b)}, \tag{66}$$

where $S_a \cap S_b = \varnothing$. Then

$$U_a U_b = U_b U_a. \tag{67}$$

This provides a precise mathematical criterion for a class of concurrent operations: commutativity follows from disjoint support, not merely from the fact that the corresponding Petri transitions are both enabled. When supports overlap, the ordering must remain explicit because non-commutativity may change the resulting state.

This distinction is important for later scalability and parallelisation studies. A structural concurrency relation in the Petri layer is not automatically equivalent to computational parallelism, and computational parallelism is not automatically equivalent to operator commutativity. The three notions must be tested separately.

18.4 Semantic consistency conditions

The preceding definitions lead to a compact consistency checklist. Structural enabling must use the declared marking. Fuzzy activation must remain in its specified domain. The encoding function must return a parameter in the domain required by the operator family. The resulting operator or channel must satisfy its mathematical admissibility conditions. The state representation must preserve the declared basis and subsystem ordering. Finally, the observation map must be applied to the state produced by the declared transformation rather than to an intermediate quantity.

These conditions provide a stronger notion of correctness than matching a single final numerical value. An implementation can produce an apparently plausible output while violating basis ordering, subsystem support, trace preservation, or the intended activation semantics. The QFLPN specification is therefore designed to make such failures individually diagnosable.

19 Semantic Non-Conflation and Identifiability of the Hybrid Interface

A central risk in hybrid mathematical models is semantic conflation: two quantities may share the same numerical interval while representing different mathematical objects. In QFLPN, the fuzzy activation λ_t, the encoded operator parameter θ_t, a quantum amplitude, and a measurement probability therefore require separate domains and interpretation rules.

19.1 Formal separation of activation and observation probability

Let $\lambda \in [0, 1]$ be a fuzzy activation and let the declared reference encoding be

$$\theta = 2 \arcsin \sqrt{\lambda}. \tag{68}$$

For an initial qubit state $|0\rangle$ and the rotation convention used in the reference instance,

$$R_y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle, \tag{69}$$

so that the probability of obtaining outcome 1 under the computational-basis measurement is

$$p_1 = \sin^2(\theta/2) = \lambda. \tag{70}$$

The equality of these two scalar values is therefore a consequence of the declared encoding and measurement convention; it is not an identification of fuzzy membership with quantum probability. Changing the encoding, the initial state, the measurement basis, or the observable can break this numerical equality while leaving the semantic distinction intact.

19.2 Counterexample to direct semantic identification

Consider $\lambda = 0.25$. Under the reference encoding, $\theta = \pi/3$, and the amplitude of $|1\rangle$ is $\sin(\pi/6) = 0.5$, whereas the corresponding measurement probability is 0.25. Thus the same transition contains at least three distinct quantities: fuzzy activation 0.25, quantum amplitude 0.5, and measurement probability 0.25. The numerical coincidence between activation and probability in this particular construction does not justify treating the quantities as interchangeable.

This counterexample exposes an implementation-level failure mode. A program that stores the fuzzy activation directly as a quantum amplitude would produce a probability of 0.0625 for the same component, whereas the declared QFLPN encoding requires 0.25. The discrepancy is not a numerical-rounding issue; it is a semantic error in the interface between the fuzzy and quantum layers.

19.3 Interface well-definedness theorem

Theorem. Let a QFLPN transition have a structural admissibility relation A_t, a fuzzy activation map g_t with range $[0, 1]$, an encoding Γ_t whose image is contained in the domain of an admissible quantum transformation Φ_t, and a declared observation map $\mathfrak{M}_t$. If the input marking satisfies A_t, the activation is computed from the declared fuzzy inputs, and the encoded parameter is evaluated before Φ_t, then the transition defines a well-typed semantic composition

$$\mathfrak{M}_t \circ \Phi_t \circ \Gamma_t \circ g_t \tag{71}$$

on its declared domain.

Proof. Structural admissibility first restricts the transition to an input marking for which g_t is defined. By construction, g_t returns an element of $[0, 1]$. The encoding Γ_t is defined on this interval and returns an element of the parameter domain required by Φ_t. Because Φ_t is admissible on that parameter domain, it maps the declared quantum-state space to itself or to the declared output state space. Finally, $\mathfrak{M}_t$ is applied only to the resulting state and therefore has a well-defined input. The composition is consequently defined on every admissible transition input. The theorem establishes semantic well-typedness; it does not establish numerical accuracy, physical fidelity, or computational efficiency.

The theorem gives P1 a stronger formal boundary: correctness can be decomposed into independently testable interface obligations. A failure in g_t, Γ_t, Φ_t, or $\mathfrak{M}_t$ can be localized without changing the interpretation of the remaining layers. This modularity is a prerequisite for meaningful comparison in later articles because a numerical improvement is scientifically interpretable only when the compared implementations instantiate the same semantic contract.

20 Scientific Novelty and Contribution Boundary

The novelty claim of P1 is intentionally narrow and testable. The paper contributes a unified operator-level specification in which discrete Petri structure, bounded fuzzy activation, and quantum state evolution are connected through explicit interfaces. The novelty is the formal integration and typing of these interfaces, together with admissibility conditions and a reproducible reference construction.

The contribution should not be stated as the invention of Petri nets, fuzzy logic, quantum mechanics, unitary evolution, CPTP maps, or tensor-product Hilbert spaces. These are established components. The scientific contribution lies in how they are assembled into one formally constrained model and in the consequences that can be proved from that assembly.

For the same reason, P1 deliberately avoids absorbing the principal empirical questions assigned to the later articles. Reachability and event-structure analysis, sparse computational scaling, numerical comparison of matrix-function methods, degradation modelling, decoherence robustness, spectral convergence, tensor-network comparison, and application-level validation are separate research questions. Keeping these boundaries explicit is part of the scientific design of the publication portfolio, not an editorial limitation.

A strong publication program therefore treats each article as an independent unit with its own research question, methodological object, evidence set, figures and tables, while allowing the thesis to synthesize the results at a higher level. Shared definitions that are mathematically unavoidable should be cited and, when reused from an earlier publication, explicitly attributed. New claims should never be presented as though they were established by an earlier paper merely because they use the same QFLPN notation.

21 Discussion

The formal contribution can be understood as a typed interface between three mathematical domains. The Petri layer controls structure and event precedence. The fuzzy layer supplies a bounded activation value. The quantum layer receives a declared operator parameter and evolves a state according to unitary or channel semantics. Because the interfaces are explicit, the layers can be refined independently without changing the meaning of the other layers.

The activation-to-rotation map illustrates this modularity. The mapping $\theta = 2 \arcsin \sqrt{\lambda}$ is useful because it gives an exact probability relation for the declared initial state and measurement convention. However, it should not be interpreted as a universal law connecting fuzzy logic to quantum mechanics. It is a modelling choice whose usefulness must be evaluated in the application for which it is selected.

The formalism also provides a natural route toward open-system extensions. Replacing $U\rho U^\dagger$ by a CPTP map allows the same transition semantics to represent noisy or dissipative evolution. A Lindblad generator can subsequently be introduced for continuous-time dynamics,

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_a \left(L_a \rho L_a^\dagger - \frac{1}{2}\{L_a^\dagger L_a, \rho\} \right), \tag{72}$$

provided that the generator and numerical integration scheme are specified. That extension belongs to the dedicated decoherence study rather than being presented here as an obtained experimental result.

A second extension concerns numerical matrix functions. Once a QFLPN transition is expressed through an operator exponential, one may study Taylor, Krylov, Arnoldi, or Lanczos approximations. The existence of the formal operator interface does not establish which approximation is fastest or most accurate. Such a comparison requires a controlled benchmark, error metric, stopping rule, hardware description, and repeated measurements.

A third extension concerns sparse state evolution. The formal tensor-product construction determines the dimension of the state space but does not determine the sparsity pattern of a particular operator. Therefore, claims about CSR efficiency or parallel speedup must be based on measured implementations. The same distinction applies to tensor-network representations: a tensor-network compression may be advantageous for some states and ineffective for others depending on entanglement structure and truncation criteria.

The framework therefore supports a coherent research portfolio while preserving a strict boundary between formal consequences and empirical results. This is particularly important for a multi-paper doctoral research program: each paper should answer a distinct research question, present its own evidence, and avoid borrowing unmeasured results from another component.

22 Limitations and Evidence Boundary

The present paper has four principal limitations. First, the fuzzy aggregation function is left application-dependent; the formalism specifies its domain and range but does not select a universally optimal fuzzy rule. Second, the reference activation-to-rotation map is one encoding among possible alternatives. Third, the four-qubit reference instance is intentionally small and cannot establish large-scale computational behavior. Fourth, the paper does not report physical hardware experiments or measured noise characteristics.

These limitations are not missing sections to be hidden by additional prose. They define the boundary of what can legitimately be concluded from P1. A scientific article becomes stronger when the evidence boundary is explicit and subsequent experiments are designed to answer the unresolved questions.

The later QFLPN studies are therefore logically separated. The four-qubit instance study can establish implementation-level reproducibility. The sparse computational study can establish measured performance for declared workloads. The matrix-function study can compare numerical algorithms. The degradation and decoherence studies can evaluate open-system effects. Spectral analysis can establish stability properties under declared assumptions. Comparative tensor-network work can assess representation-dependent trade-offs. An application study can then integrate validated components.

23 Conclusion

A formal operator framework for Quantum-Fuzzy Logical Petri Nets has been specified by separating discrete Petri structure, fuzzy activation, and quantum-state evolution into explicit semantic layers. The QFLPN tuple, local and global quantum markings, bounded activation, activation-to-operator mapping, transition semantics, and unitary or CPTP evolution provide a coherent mathematical interface between the three components.

The framework supplies direct analytical verification conditions for density operators and unitary evolution and admits a deterministic four-qubit reference with fixed dimension, basis ordering, activation encoding, and controlled-operation conventions. The reference construction provides a concrete anchor for implementation and independent verification.

The principal scientific boundary is explicit: the formalism does not by itself establish scalability, numerical superiority, physical calibration, or hardware robustness. Those claims require separate evidence. Preserving this distinction allows the subsequent QFLPN studies to contribute independent and verifiable results without retroactively attributing empirical properties to the formal definitions.

24 Data and Code Availability

A complete reproducibility package for the reference instance should contain the exact QFLPN specification, basis-order convention, operator-construction code, numerical precision, input activations, initial state, measurement convention, verification tolerances, software environment, and generated outputs. The final package should be archived together with the article-specific code and, where applicable, the benchmark data used in subsequent performance studies. The present paper defines the required structure of that package; a claim of independent execution should be made only after the corresponding artifacts have been generated and checked.

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